

Lesson 1 : Tools in the Workshop

At the beginning of the technical English course it is necessary to introduce some basic vocabulary and grammar. Technical English is different from the English taught at secondary schools. For this reason, the following section provides the basic tools for every student of mechanical engineering to enlarge their knowledge and to work in an international surrounding.

The first and important thing to know is what all the tools and machines in the workshop are called. This enables an engineer to give precise instructions to the trained workers about what to do and how. Therefore, the exercises will provide an overview of the content of a toolbox, useful verbs, measurement tools and the most common units that are needed in during daily working processes.

When it comes to measuring work pieces, there are various methods and many tools available for this task. The first differentiation has to be made between gauges and measurement equipment.

The first represent either a measurement or a form that refers to limit dimensions of tolerances. These have to be fulfilled in fits and usually just provide information about whether a fit is within the limit of tolerances or not. This group includes gauging tools, e.g : slip gauges and accidences, straightedge, square and limit gauges, cylindrical plug gauges, gauging rings and calliper gauges.

The latter can be used for acquiring information on the measurements of a work piece and provide information on the length, width and depth of the outer or inner edges of a work piece, hole or slot. The most common instruments in mechanical engineering are vernier callipers or calliper rules which can be used universally in the workshop. They provide information on lengths and are precise to 0.1 mm. If it comes to smaller tolerances, micrometers

are usually used. They measure accurately to within 0.001 mm. Com